Practice Quiz: Communication (Neuroscientific Frontiers)

Name: __ ***Date:*** __

Part A: Multiple Choice

1. The right TPJ is critical for: A) Motor control B) Self-other distinction — maintaining separate generative models for self and other, enabling theory of mind C) Visual processing D) Auditory processing

2. Active Inference reinterprets mirror neurons as: A) Special matching neurons B) Ordinary neurons in a predictive model that generates the same predictions for self-action and observed-action C) Exclusively motor neurons D) Inhibitory interneurons

3. The N400 ERP component represents: A) Motor preparation B) A lexical-semantic prediction error — the brain's surprise at an unexpected word in context C) Sleep onset D) Pain

4. Interpersonal neural coupling during conversation reflects: A) Electrical interference B) Mutual model alignment — both brains generating similar predictions, producing correlated neural activity C) Hearing the same sounds D) Eye contact

5. Oxytocin functions in Active Inference as: A) A sleep hormone B) A social precision modulator — increasing the gain (weight) of social prediction errors C) A motor neurotransmitter D) A memory enhancer

6. The STS specializes in: A) Taste B) Inferring action intentions from biological motion — reading body language, gaze, and gesture through predictive models C) Smell D) Balance

7. Language prediction in the brain occurs at: A) Only the word level B) Multiple hierarchical levels simultaneously — phonological, lexical, syntactic, and discourse C) Only the sentence level D) Only after the sentence is complete

Part B: Essay Questions

1. Compare the neural basis of understanding a friend's emotional story (social inference) with understanding a news article (language inference). Both involve predictive models, but they recruit partially different networks. Map the overlapping and distinct neural contributions, explaining what each brain region adds to comprehension. (400 words)

2. Autism has been proposed as a condition of altered social precision — either too-high precision on sensory prediction errors (weak central coherence) or too-low precision on social priors (reduced social prediction). Review the neural evidence for each proposal. How does each account explain: (a) difficulties with eye contact, (b) reduced theory of mind performance, (c) enhanced detail perception, and (d) predictable routine preferences? (400 words)

3. Design a study to test whether interbrain synchronization during therapy predicts therapeutic outcomes. Specify the therapeutic context, neural measurements, analysis approach, and what finding would support Active Inference's account of therapeutic communication as mutual model alignment. (400 words)